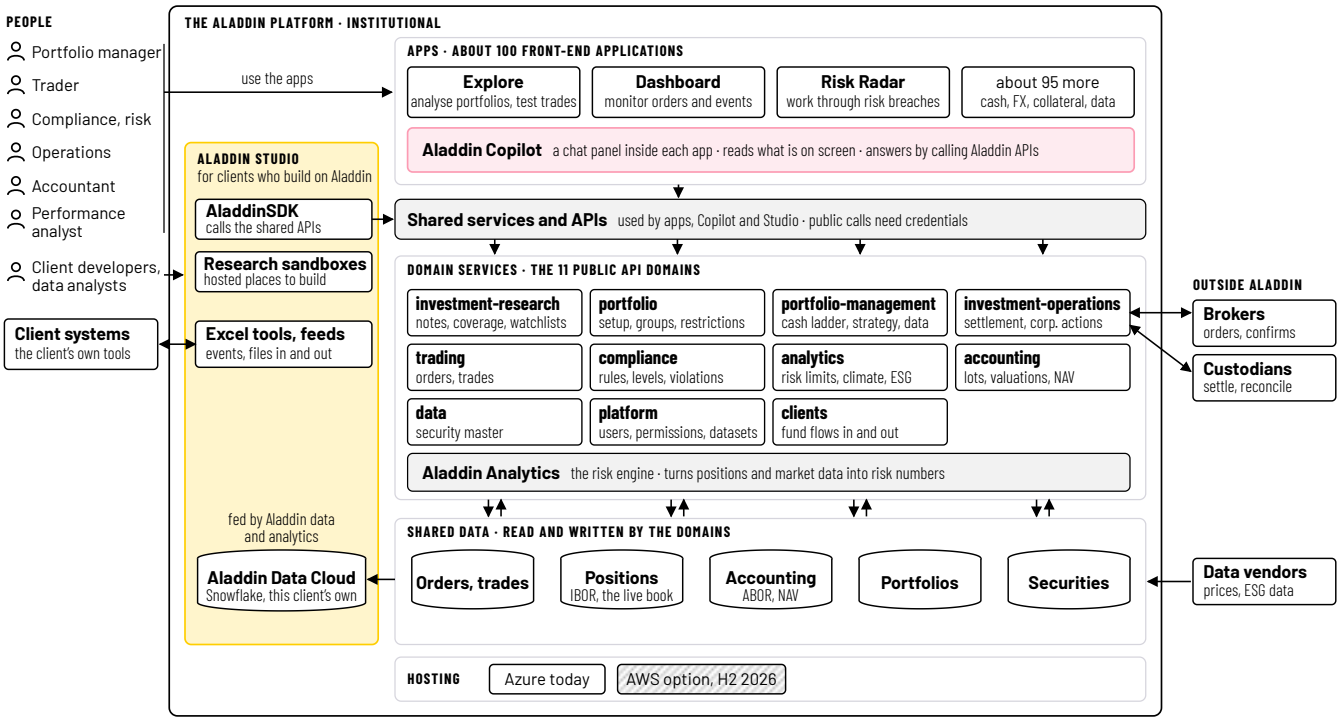
Exhibit 3. Apps, Copilot and Studio share the same Aladdin services and data



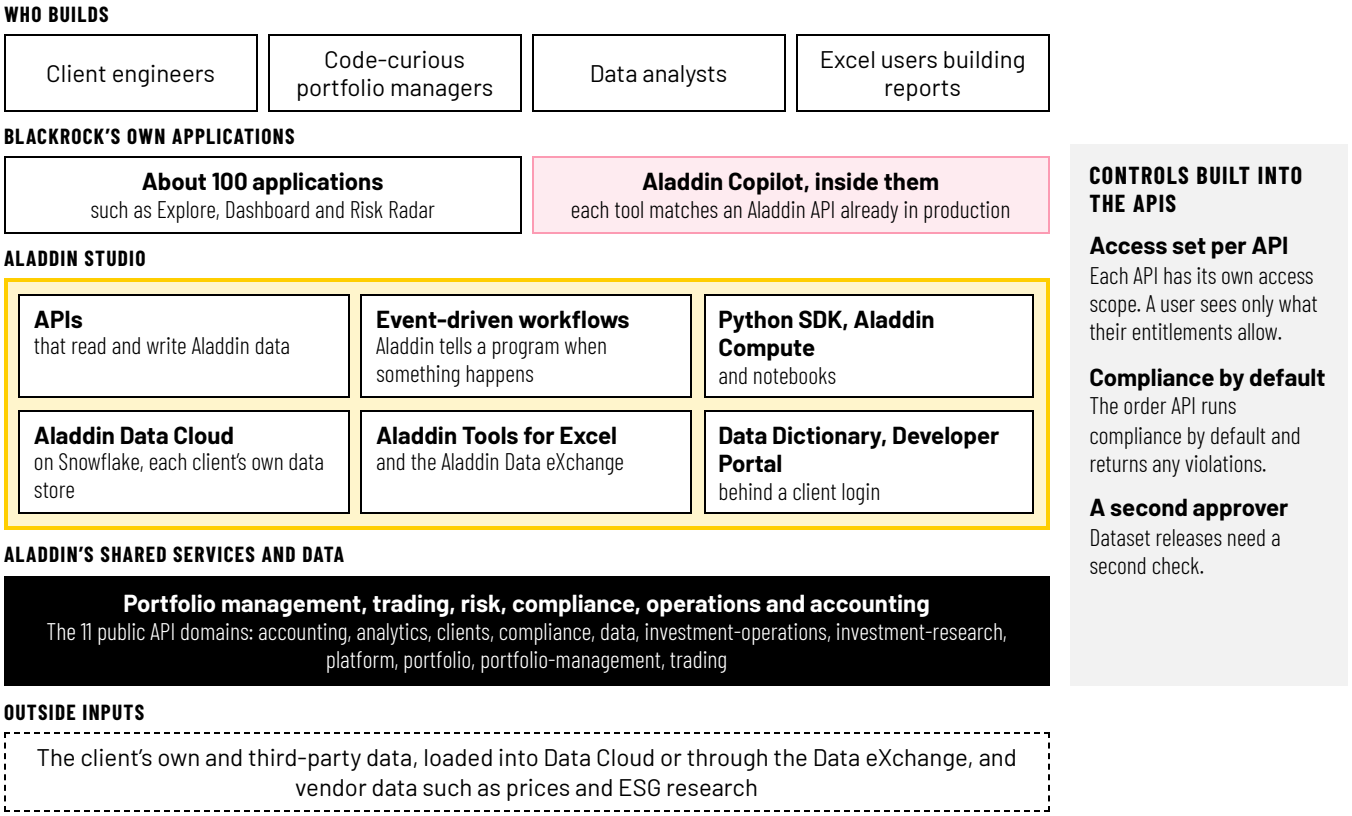
For illustrative purposes only. The 11 domains are the packages of BlackRock's public SDK, named as in its specifications. Other app names come from those specifications, so their screens are not drawn. ■ Source: Rosales and Vicente Valdez at 2:35, for the apps and Copilot[9]; Woodhead at 4:49, for Studio, events and the Data Cloud[1]; the public specifications and SDK[6, 29]; a training recording, for Explore[27]; BlackRock's Studio, APIs, ATX, ADX, Data Cloud and Aladdin Risk pages[14, 20, 30, 31, 32, 33]; the Data Cloud launch release[34]; hosting[21, 35]; data vendors, ICE and Sustainalytics[36, 37]. Solid outline = shown in sources; hatched = announced, not live.

Exhibit 4. Studio, grouped here into six pieces

Piece	What it is, in plain words
APIs (Aladdin Graph)	Ways for a program to read and write Aladdin data. Aladdin Graph[29, 38] is the whole API layer, not one API. Every public specification's id starts `agraph.<domain>`[6].
Event-driven workflows	Aladdin tells a program when something happens[31, 39]. Woodhead names "APIs events" as one part[1]; Diwakar Goel, Global Head of Data, Aladdin, wants "every single transaction … via an event, via an API"[40].
Aladdin Data Cloud	A managed data store on Snowflake, with Fivetran moving the data[14, 28]. Goel puts its latency at "within 15 minutes"[40].
Aladdin Compute	A hosted analytics platform with ready-to-use Python environments[41]. The SDK has a run mode for "Aladdin Studio Compute sessions"[29].
Aladdin SDK	An open-source Python library that handles login, retries, rate limits, pagination and long-running operations[29]. Eleven add-on packages, one per API domain, are generated from the specifications by BlackRock's plugin builder[42].
Developer Portal and Data Dictionary	The API catalog, framework guides, field dictionary, notebooks and code samples the APIs page describes[30]. A BlackRock job posting names the "Aladdin Studio Developer Portal"[39].

■ Source: BlackRock Studio and APIs pages[14, 30]; the 54 public specifications and SDK README[6, 29]; Woodhead[1], Goel[40] and Fivetran[28]. BlackRock gives no count. Its APIs page also links Aladdin Data eXchange and Aladdin Tools for Excel, which Exhibit 3 shows[30]. No event specification is public[6]. The APIs page describes the catalog, guides, notebooks and samples but does not link them[30]. The specifications link the guides to a portal behind a client login[6, 43].

Exhibit 3. Apps, Copilot and Studio share the same Aladdin services and data



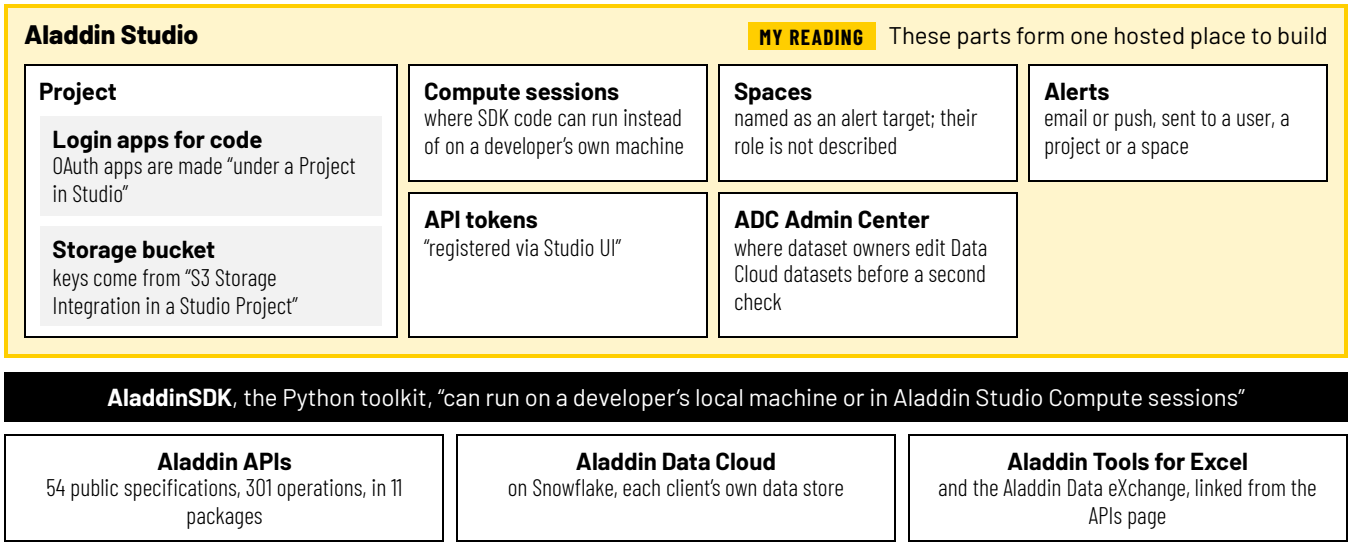
For illustrative purposes only. The 11 domains are the packages of BlackRock's public SDK, named as in its specifications, which also name Dashboard and Risk Radar. The controls are those in Exhibit 14. ■ Source: Rosales and Vicente Valdez at 2:35, for the apps and Copilot[9]; a training recording, for Explore[27]; Woodhead at 4:49, for Studio, events and the Data Cloud[1]; the public specifications and SDK[6, 29]; BlackRock's Studio, APIs, ATX, ADX, Data Cloud and Aladdin Risk pages[14, 20, 30, 31, 32, 33]; the developer portal[34]; the Data Cloud launch release[35]; data vendors, ICE and Sustainalytics[36, 37]. Aladdin is hosted on Microsoft Azure, with an AWS option announced for US-hosted Aladdin Enterprise clients[21, 38].

Exhibit 4. Studio, grouped here into six pieces

Piece	What it is, in plain words
APIs (Aladdin Graph)	Ways for a program to read and write Aladdin data. Aladdin Graph[29, 39] is the whole API layer, not one API. Every public specification's id starts `agraph.<domain>`[6].
Event-driven workflows	Aladdin tells a program when something happens[31, 40]. Woodhead names "APIs events" as one part[1]; Diwakar Goel, Global Head of Data, Aladdin, wants "every single transaction … via an event, via an API"[41].
Aladdin Data Cloud	A managed data store on Snowflake, with Fivetran moving the data[14, 28]. Goel puts its latency at "within 15 minutes"[41].
Aladdin Compute	A hosted analytics platform with ready-to-use Python environments[42]. The SDK has a run mode for "Aladdin Studio Compute sessions"[29].
Aladdin SDK	An open-source Python library that handles login, retries, rate limits, pagination and long-running operations[29]. Eleven add-on packages, one per API domain, are generated from the specifications by BlackRock's plugin builder[43].
Developer Portal and Data Dictionary	The API catalog, framework guides, field dictionary, notebooks and code samples the APIs page describes[30]. A BlackRock job posting names the "Aladdin Studio Developer Portal"[40].

■ Source: BlackRock Studio and APIs pages[14, 30]; the 54 public specifications and SDK README[6, 29]; Woodhead[1], Goel[41] and Fivetran[28]. BlackRock gives no count. Its APIs page also links Aladdin Data eXchange and Aladdin Tools for Excel, which Exhibit 3 shows[30]. No event specification is public[6]. The APIs page describes the catalog, guides, notebooks and samples but does not link them[30]. The specifications link the guides to a portal behind a client login[6, 34].

Exhibit 3. The Studio parts named in BlackRock's public SDK files



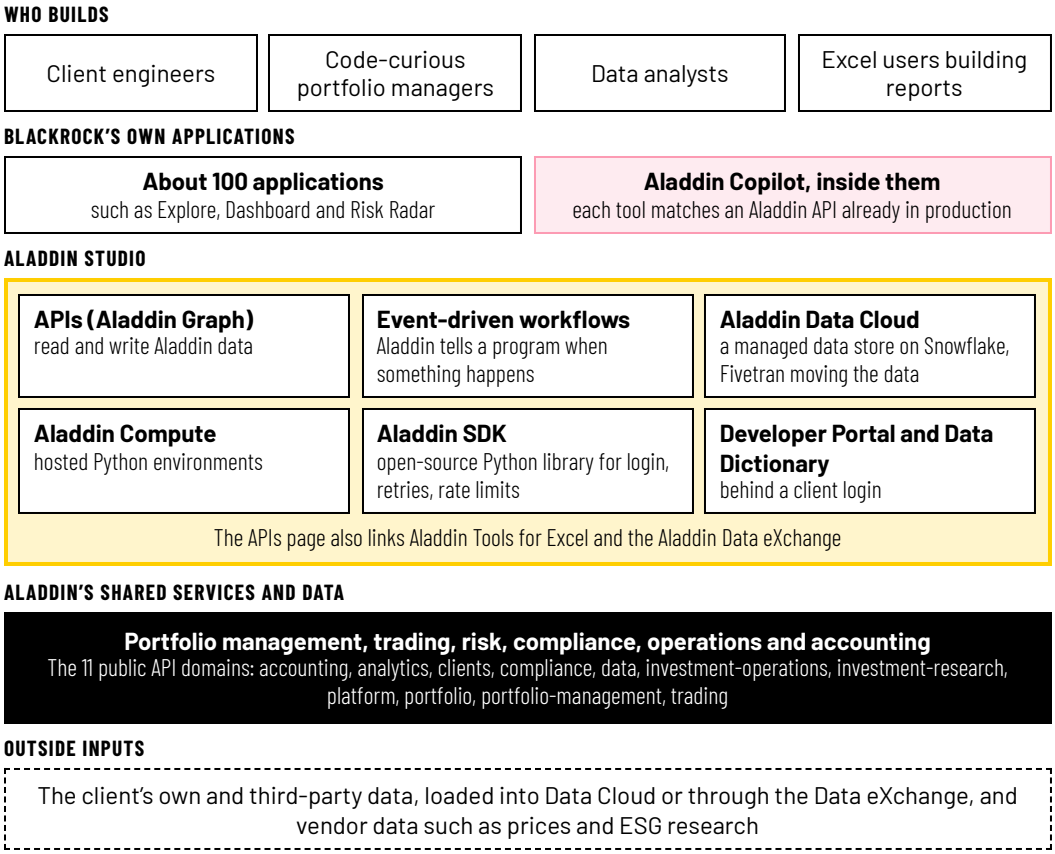
For illustrative purposes only. Every part inside the Studio box is named in the SDK README's settings for login, storage, API tokens, run mode and notifications, or in the Data Cloud dataset specification. The files do not say what a Space is. Around these parts sit BlackRock's own applications and Copilot, which draw on the same Aladdin services and data[9], among them Explore[27] and the risk engine[29]; Aladdin is hosted on Microsoft Azure, with an AWS option announced for US-hosted Aladdin Enterprise clients[21, 30], and takes in vendor data such as ICE prices and Sustainalytics ESG research[31, 32]. ■ Source: the SDK README[33]; the public specifications[6]; Woodhead at 4:49, for Studio and the Data Cloud[1]; BlackRock's Studio, APIs, ATX, ADX and Data Cloud pages[14, 20, 34, 35, 36]; the Data Cloud launch release[37].

Exhibit 4. Studio, grouped here into six pieces

Piece	What it is, in plain words
APIs (Aladdin Graph)	Ways for a program to read and write Aladdin data. Aladdin Graph[33, 38] is the whole API layer, not one API. Every public specification's id starts `agraph.<domain>`[6].
Event-driven workflows	Aladdin tells a program when something happens[29, 39]. Woodhead names "APIs events" as one part[1]; Diwakar Goel, Global Head of Data, Aladdin, wants "every single transaction … via an event, via an API"[40].
Aladdin Data Cloud	A managed data store on Snowflake, with Fivetran moving the data[14, 28]. Goel puts its latency at "within 15 minutes"[40].
Aladdin Compute	A hosted analytics platform with ready-to-use Python environments[41]. The SDK has a run mode for "Aladdin Studio Compute sessions"[33].
Aladdin SDK	An open-source Python library that handles login, retries, rate limits, pagination and long-running operations[33]. Eleven add-on packages, one per API domain, are generated from the specifications by BlackRock's plugin builder[42].
Developer Portal and Data Dictionary	The API catalog, framework guides, field dictionary, notebooks and code samples the APIs page describes[34]. A BlackRock job posting names the "Aladdin Studio Developer Portal"[39].

■ Source: BlackRock Studio and APIs pages[14, 34]; the 54 public specifications and SDK README[6, 33]; Woodhead[1], Goel[40] and Fivetran[28]. BlackRock gives no count. Its APIs page also links Aladdin Data eXchange and Aladdin Tools for Excel, which Exhibit 3 shows[34]. No event specification is public[6]. The APIs page describes the catalog, guides, notebooks and samples but does not link them[34]. The specifications link the guides to a portal behind a client login[6, 43].

Exhibit 3. Apps, Copilot and Studio share the same Aladdin services and data



CONTROLS BUILT INTO THE APIS

Access set per API
Each API has its own access scope. A user sees only what their entitlements allow.

Compliance by default
The order API runs compliance by default and returns any violations.

A second approver
Dataset releases need a second check.

For illustrative purposes only. The six Studio pieces are my grouping; BlackRock gives no count. The 11 domains are the packages of BlackRock's public SDK, named as in its specifications, which also name Dashboard and Risk Radar. No event specification is public, and the specifications link the guides to a portal behind a client login. The controls are those in Exhibit 13. ■ Source: Rosales and Vicente Valdez at 2:35, for the apps and Copilot[9]; a training recording, for Explore[27]; Woodhead at 4:49, for Studio, events and the Data Cloud[1]; the 54 public specifications and the SDK README[6, 29]; Naik and Kalish, for Aladdin Graph[30]; the plugin builder[31]; Snowflake's partner page, for Compute[32]; Fivetran[28]; Goel, for events[33]; a BlackRock job posting[34]; BlackRock's Studio, APIs, ATX, ADX, Data Cloud and Aladdin Risk pages[14, 20, 35, 36, 37, 38]; the developer portal[39]; the Data Cloud launch release[40]; data vendors, ICE and Sustainalytics[41, 42]. Aladdin is hosted on Microsoft Azure, with an AWS option announced for US-hosted Aladdin Enterprise clients[21, 43].

Exhibit 4. Two domains hold 127 of the 301 public API operations

Domain	Specs	Operations	Alpha	Beta	Stable	Sunset	Largest APIs by operations
Investment operations	15	**74**	9	4	2	–	PrincipalInterestFactor, CounterPartySettlementInstruction, MiscellaneousCashflow, CouponReset and 11 more
Analytics	9	**53**	–	–	8	1	RiskRule, RiskWorkflow, RiskException, RiskConfig and 5 more
Investment research	5	**41**	5	–	–	–	AnalystCoverage, Criterion, Watchlist, ResearchNote and 1 more
Platform	6	**29**	2	–	4	–	AdcDataset, UserGroup, User, UserGroupPermission and 2 more
Portfolio management	4	**28**	1	–	3	–	CIDWriter, Strategy, CIDReader, CashLadder
Portfolio	3	**23**	1	–	2	–	PortfolioGroup, RestrictedAsset, PortfolioToolkit
Data	1	**15**	–	–	1	–	SecurityCreation
Trading	2	**14**	1	–	1	–	Order, Trade
Compliance	4	**9**	3	–	1	–	ComplianceRuleAssignment, Violation, ComplianceRule, Level

Domain	Specs	Operations	Alpha	Beta	Stable	Sunset	Largest APIs by operations
Accounting	4	**8**	2	2	–	–	PortfolioConfiguration, Valuations, CompositeMembership, AborLot
Clients	1	**7**	–	–	1	–	EnrichedCapitalFlow
Total	**54**	**301**	**24**	**6**	**23**	**1**	11 domains, 11 PyPI packages

Source: the 54 Swagger 2.0 specifications inside the 11 `asdk_plugin` packages, version 2.0.0b3, on PyPI[6]. Counts are mine, recomputed from the files. The core `aladdinsdk` package adds four utility APIs (token, subscription, notification and a demo), not counted here. Alpha, beta, stable and sunset are the stability labels each specification carries (its `x-aladdin-api-stability` field), alpha being the earliest; the four columns count specs by label.

The specifications say more than Exhibit 4's counts: they show how the APIs behave. Posting an order takes `whatIf` and `runCompliance`, which defaults to true, and the response returns any compliance violations inline[6]. Ten specifications expose an endpoint for long-running operations, and 35 operations are flagged as long-running[6]. The SDK polls them[29]. Risk exceptions and Data Cloud datasets carry bitemporal fields, which record two time dimensions. Rule assignments and dataset releases name a four-eyes approval, a second sign-off[6]. Every request carries an application key and a second credential. That is an OAuth 2.0 token, a client key, or a user name and password[6]. OAuth scopes look like `trading.order_management.order.v1.Order:read`[6]. The client ID and secret come from the Studio UI[29].

What Studio is

I pulled about 125 recordings. Five BlackRock speakers in them describe Studio or its parts[1, 22, 33, 44, 45]. Exhibit 5 lists each with the title the recording gives. BlackRock's own definition comes from the head of Studio engineering, on Snowflake's podcast in November 2025.

> "So we think about Aladdin today as a platform and Aladdin Studio is essentially the user experience that allows people to think of Aladdin as a platform. It exposes everything Aladdin knows how to do as a set of APIs events. The Aladdin data cloud built on top of Snowflake takes all of that information that's generated through our analytics factory and inside of Aladdin and pushes that into Snowflake. And increasingly we're building a set of developer tools."
>
> — Dave Woodhead, Managing Director and Head of Engineering for Aladdin Studio, Snowflake Data Cloud podcast, published November 11, 2025[1]

Exhibit 5. Five BlackRock executives describe Studio or its parts on the record

Speaker	Title, as the recording gives it	What they add to the definition	Recording
Dave Woodhead	Managing Director and Head of Engineering for Aladdin Studio	Gives the three-part definition quoted above	Snowflake podcast, November 2025[1]
Richard Newman	General Manager, Aladdin Studio	Data Cloud is "part of Studio"; analytics APIs "releasing soon … all part of the Studio umbrella"	Never Done podcast, January 2025[22]
Diwakar Goel	Global Head of Data, Aladdin	Every transaction "via an event, via an API"; Data Cloud "within 15 minutes of latency"	Pensions webinar, 2025[33]